STAT 405 - Project Report

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Data

We use publicly available crime data from the Vancouver Police Department, containing yearly counts of multiple crime categories across neighbourhoods in Vancouver.

Since the data were provided in PDF format, we manually constructed a dataset in R. Each observation y_{ij} represents the count of crime type j in neighbourhood i.

```r
crime_data <- data.frame(
  neighbourhood = c(
    "Arbutus Ridge","Central Business District","Dunbar-Southlands","Fairview",
    "Grandview-Woodland","Hastings-Sunrise","Kensington-Cedar Cottage",
    "Kerrisdale","Killarney","Kitsilano","Marpole","Mount Pleasant","Musqueam",
    "Oakridge","Renfrew-Collingwood","Riley Park","Shaughnessy","South Cambie",
    "Stanley Park","Strathcona","Sunset","Victoria-Fraserview","West End",
    "West Point Grey"),
  sex_offences = c( 0,151,8,16,27,29,34,6,14,12,8,24,2,8,30,12,0,13,1,43,24,15,43,4),
  assaults = c(18,1591,22,115,254,97,185,22,51,101,69,215,15,26,168,41,15,24,18,622,80,
               45,306,24),
  robbery = c( 1,218,0,9,22,7,26,3,3,10,4,36,0,0,22,3,2,2,4,64,9,9,35,2),
  break_enter = c(25,438,24,164,121,84,127,29,37,92,89,173,0,44,123,75,32,33,12,
                  151,77,29,149,29),
  theft_mv = c(7,74,7,16,64,31,48,10,24,18,22,28,0,7,53,13,6,6,2,48,36,24,23,5),
  theft_auto = c(41,1855,72,219,219,222,317,73,84,202,146,220,0,56,202,148,43,
                 56,54,369,148,136,443,60),
  theft_under_5k = c(69,4458,70,706,507,312,601,132,159,465,276,1015,
                     1,115,1446,178,38,140,53,603,637,125,1120,77),
  arson = c(1,68,1,6,15,9,8,0,1,9,1,20,0,0,8,5,0,2,2,37,4,3,10,0),
  mischief = c(34,1864,33,209,281,140,209,45,83,142,134,296,6,29,171,86,28,42,16,434,137,
               85,318,32),
  offensive_weapons = c(4,708,4,28,44,9,25,5,5,10,8,39,0,1,9,2,3,4,4,244,9,10,38,3))
```

```r
head(crime_data)
```

```
##               neighbourhood sex_offences assaults robbery break_enter theft_mv
## 1             Arbutus Ridge            0       18       1          25        7
## 2 Central Business District          151     1591     218         438       74
## 3         Dunbar-Southlands            8       22       0          24        7
## 4                  Fairview           16      115       9         164       16
## 5        Grandview-Woodland           27      254      22         121       64
## 6          Hastings-Sunrise           29       97       7          84       31
##   theft_auto theft_under_5k arson mischief offensive_weapons
## 1         41             69     1       34                 4
## 2       1855           4458    68     1864               708
## 3         72             70     1       33                 4
## 4        219            706     6      209                28
```

```
## 5          219              507     15       281                  44
## 6          222              312      9       140                   9
```

Models

Models

We model crime counts y_{ij} using Poisson distributions.

Model 1 (Independent model):

$$y_{ij} \mid \lambda_{ij} \sim \text{Poisson}(\lambda_{ij}), \quad \lambda_{ij} \sim \text{Gamma}(\alpha, \beta)$$

Model 2 (Hierarchical model):

$$y_{ij} \mid \lambda_{ij} \sim \text{Poisson}(\lambda_{ij}), \quad \log(\lambda_{ij}) = \mu + \alpha_i + \beta_j$$

$$\alpha_i \sim \mathcal{N}(0, \sigma_\alpha^2), \quad \beta_j \sim \mathcal{N}(0, \sigma_\beta^2)$$

We impose $\sum_i \alpha_i = 0$ and $\sum_j \beta_j = 0$ for identifiability.

We interpret λ_{ij} as the expected number of crimes per year for crime type j in neighbourhood i, representing a yearly rate without explicit adjustment for exposure or opportunity.

Questions

Using the `crime_data` and specified models above we hope to answer the following questions using bayesian analysis:

- What are the underlying rates of different categories of crime across Vancouver neighbourhoods?
- What future crime counts might be predicted for selected neighbourhood and crime-type combinations?

Posterior Laplace Approximation

For Model 1, the posterior distribution can be computed exactly because the Gamma prior is conjugate to the Poisson likelihood.

```r
alpha <- 1
beta <- 1
Y <- as.matrix(crime_data[, -1])
post_shape <- alpha + Y
post_rate  <- beta + 1
post_mean <- post_shape / post_rate
post_var  <- post_shape / (post_rate^2)

# 95% credible intervals
post_lower <- qgamma(0.025, shape = post_shape, rate = post_rate)
post_upper <- qgamma(0.975, shape = post_shape, rate = post_rate)
model1_results <- data.frame(
  neighbourhood = rep(crime_data$neighbourhood, times = ncol(Y)),
  crime_type = rep(colnames(Y), each = nrow(Y)),
  observed_count = as.vector(Y),
  post_shape = as.vector(post_shape),
  post_rate = as.vector(post_rate),
  posterior_mean = as.vector(post_mean),
  posterior_variance = as.vector(post_var),
```

```
  ci_lower = as.vector(post_lower),
  ci_upper = as.vector(post_upper)
)
```

In contrast, for Model 2 the posterior does not have a closed-form expression, since the log-linear hierarchical structure introduces dependence through the neighbourhood and crime-type effects. Therefore, we use a Laplace approximation to obtain an approximate posterior distribution.

```
Y <- as.matrix(crime_data[, -1])
rownames(Y) <- crime_data$neighbourhood
I <- nrow(Y)
J <- ncol(Y)
# par = (mu, alpha_1,...,alpha_{I-1}, beta_1,...,beta_{J-1})
log_posterior <- function(par, Y, sigma_alpha = 1, sigma_beta = 1) {
  mu <- par[1]
  alpha_free <- par[2:I]
  beta_free  <- par[(I + 1):(I + J - 1)]
  alpha <- c(alpha_free, -sum(alpha_free))
  beta  <- c(beta_free,  -sum(beta_free))
  eta <- outer(alpha, beta, "+") + mu
  loglik <- sum(Y * eta - exp(eta))
  logprior <- -sum(alpha^2) / (2 * sigma_alpha^2) - sum(beta^2)  / (2 * sigma_beta^2)
  loglik + logprior
}
```

```
p <- 1 + (I - 1) + (J - 1)
init <- rep(0, p)
opt <- optim(par = init,
  fn = log_posterior,
  Y = Y,
  sigma_alpha = 1,
  sigma_beta = 1,
  method = "BFGS",
  control = list(fnscale = -1, maxit = 50000, reltol = 1e-10),
  hessian = TRUE)
theta_hat <- opt$par
H_hat <- opt$hessian
Sigma_hat <- solve(-H_hat)
se_hat <- sqrt(diag(Sigma_hat))
```

We compute the MAP estimate $\hat{\theta}$ using `optim()` and approximate the posterior as

$$\theta \mid y \approx \mathcal{N}\Big(\hat{\theta},\, (-H(\hat{\theta}))^{-1}\Big),$$

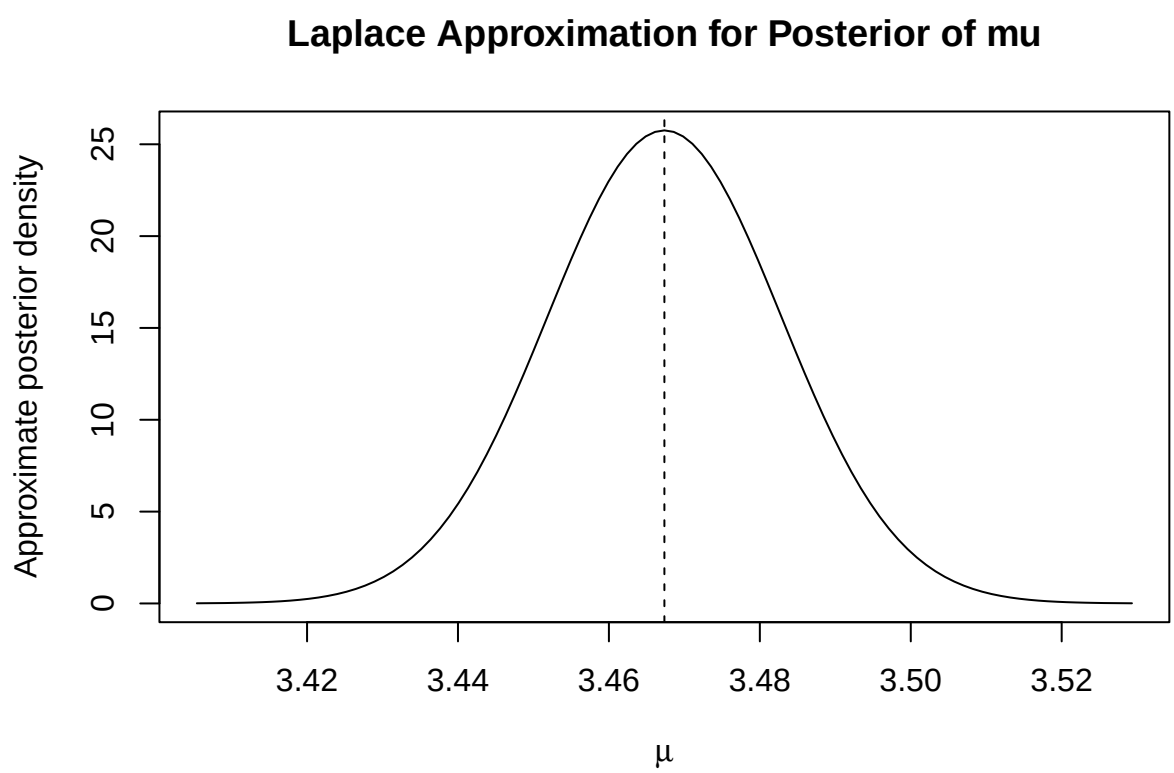
where $H(\hat{\theta})$ is the Hessian of the log-posterior.

A plot of posterior calculated is produced by code below.

```
# posterior plot for mu under Laplace approximation
mu_hat <- theta_hat[1]
mu_se  <- se_hat[1]

curve(
  dnorm(x, mean = mu_hat, sd = mu_se),
  from = mu_hat - 4 * mu_se,
  to   = mu_hat + 4 * mu_se,
```

```r
  xlab = expression(mu),
  ylab = "Approximate posterior density",
  main = "Laplace Approximation for Posterior of mu"
)
abline(v = mu_hat, lty = 2)
```



The Laplace approximation is most accurate when the posterior is approximately Gaussian near its mode.

Posterior Results and Interpretation

We now compare the posterior results from Model 1 and Model 2 in order to answer our research questions.

Under Model 1, each (i, j) pair is treated independently, and the posterior distributions for λ_{ij} are available in closed form. The posterior means vary widely across neighbourhoods and crime types, reflecting the observed counts. For example, low-count categories (e.g., $y_{ij} = 0$) produce small but nonzero posterior means due to the influence of the prior, while high-count areas such as the Central Business District yield large posterior estimates. However, because Model 1 does not share information across groups, these estimates can be highly variable, particularly for rare events.

In Model 2, we introduce a hierarchical structure

$$\log(\lambda_{ij}) = \mu + \alpha_i + \beta_j,$$

and approximate the posterior using a Laplace approximation. The parameter μ represents the overall average log-crime intensity across all neighbourhoods and crime types. From our results, we obtain $\hat{\mu} \approx 3.47$, corresponding to an average crime rate of

$$\exp(3.47) \approx 32$$

incidents per year. This can be interpreted as a baseline level of crime, with α_i and β_j capturing deviations for specific neighbourhoods and crime categories.

Unlike Model 1, Model 2 allows for partial pooling, meaning that estimates are shrunk toward this overall mean. As a result, extreme values are moderated, and estimates for low-count categories become more stable. This produces more reliable estimates of the underlying crime rates across Vancouver.

To answer our first question, we find that crime rates differ substantially across both neighbourhoods and crime types. High-density areas such as the Central Business District exhibit consistently higher rates across multiple categories, while quieter neighbourhoods show lower overall rates. Model 2 provides a clearer picture of these differences by separating overall trends (μ) from neighbourhood and crime-specific effects.

To answer our second question, we use the posterior distributions to predict future crime counts. Under both models, the expected future count for a given (i, j) is approximately equal to the posterior mean of λ_{ij}. This implies that areas with historically high crime counts are expected to remain high, while areas with low counts remain low, though with greater uncertainty. Model 2 improves these predictions by incorporating shared structure, leading to more stable and realistic forecasts, particularly in cases with limited data.

Overall, Model 1 provides a simple baseline with exact posterior computation, while Model 2 offers a more flexible and interpretable framework that captures shared patterns across neighbourhoods and crime types. The Laplace approximation allows us to efficiently perform inference in this more complex model while still obtaining meaningful insights about crime patterns and future risk.